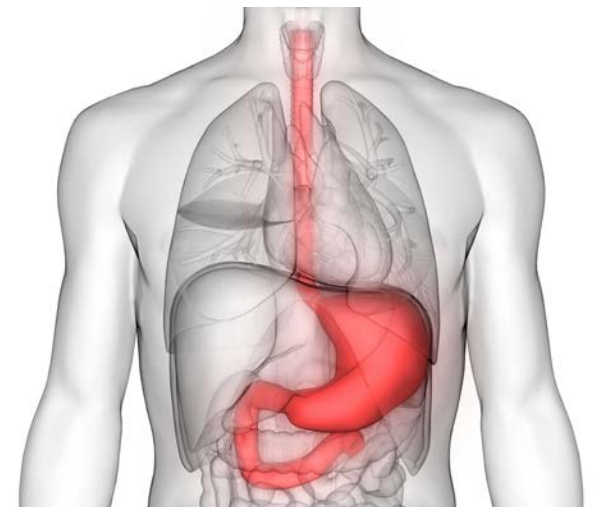
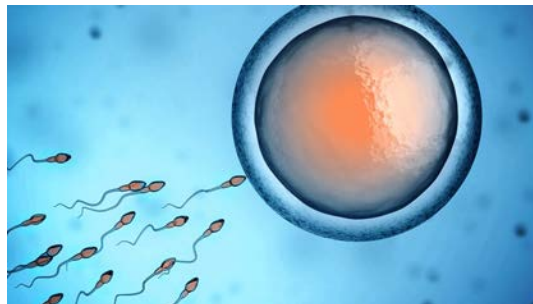
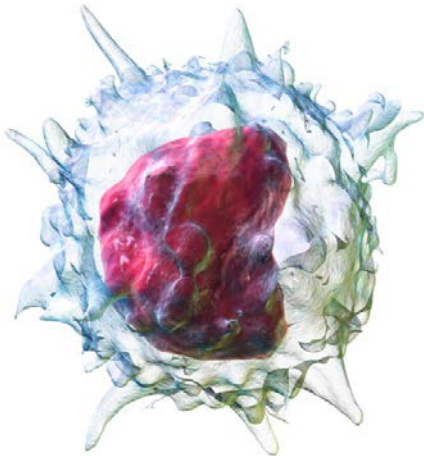
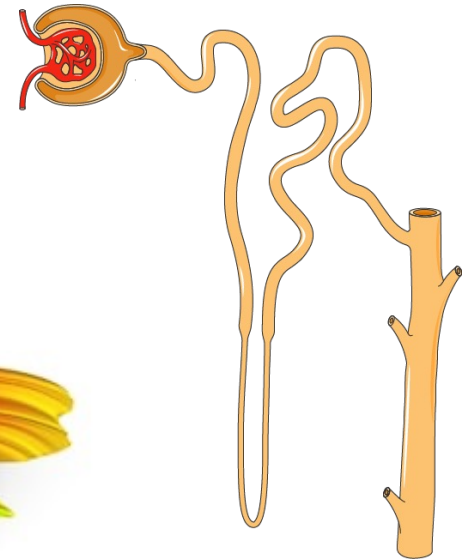
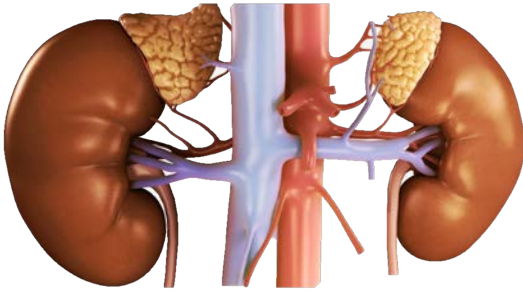


Welcome to Bio 40C!

Jeff Schinske



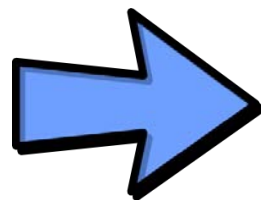


Bio 40C Proposed Norms

- Seek to enjoy your colleagues and your experiences in class!
- Actively select language that is inclusive, respectful, and welcoming
- Assume good intentions...no heroes and no villains (in this class)
- Recognize that we all have strengths as well as areas for growth
- Be curious, be skeptical
- Ask for help (and expect to receive it!)

Share with a partner...

1. Name
2. Something important to know about you as a person
3. What is something you're looking forward to this week?



Person with 1st name closest to start of alphabet goes first



Debriefing Last Week

- Last Week's Canvas Work
 - Historical kidney research context was interesting.
 - Making urine is complex!
 - Graphs! Osmosis! Oh my!!!
- Ask for help! (and expect to receive it!)

High Blood Pressure (Hypertension) and Kidney Failure



How does high blood pressure impact kidney function?

In this context, why is it so important for us to be able to accurately measure GFR?

Use the following terms in your response:

Afferent arteriole

Proteins

Kidney Failure

"High blood pressure causes the afferent arteriole to constrict, which lowers GFR lowers blood flow to reduce blood pressure in glomerulus. Continued high blood pressure damages filtration membranes, so proteins might show up in the urine. The damage can lead to fibrosis, which could reduce or stop filtration, and lead to kidney failure."

Doctors often treat chronic high blood pressure by prescribing drugs called Angiotensin Converting Enzyme (ACE) Inhibitors. These drugs help lower blood pressure and maintain proper kidney function.

Recall that, under normal conditions...

- ACE converts Angiotensin 1 to Angiotensin 2
- Angiotensin 2 constricts the efferent arteriol and constricts the mesangial cells
- On your model, constricting the efferent arteriol is like bending that part of the tube
- On your model, constricting the mesangial cells is like covering up one or more of the small holes in the glomerulus part of the tube



ACE inhibitors slow down or inhibit the activities of ACE

Based on the above information, and what you can discern from your model, explain how you think ACE Inhibitors can help avoid kidney failure.

Discuss the following four terms in your response

- | | |
|-----------------------|---|
| 1) ACE | <i>"Angiotensin 2 is responsible for signaling mesangial cells to constrict and reduce the filtration area. Angiotensin 2 also constricts the efferent arterioles, which raises blood pressure in the glomerulus. This will increase stress on the glomerulus. ACE inhibitors will inhibit the production of angiotensin 2, the mesangial cells and efferent arterioles are never signaled, so blood pressure is kept low and GFR is</i> |
| 2) angiotensin | |
| 3) mesangial cells | |
| 4) efferent arteriole | |

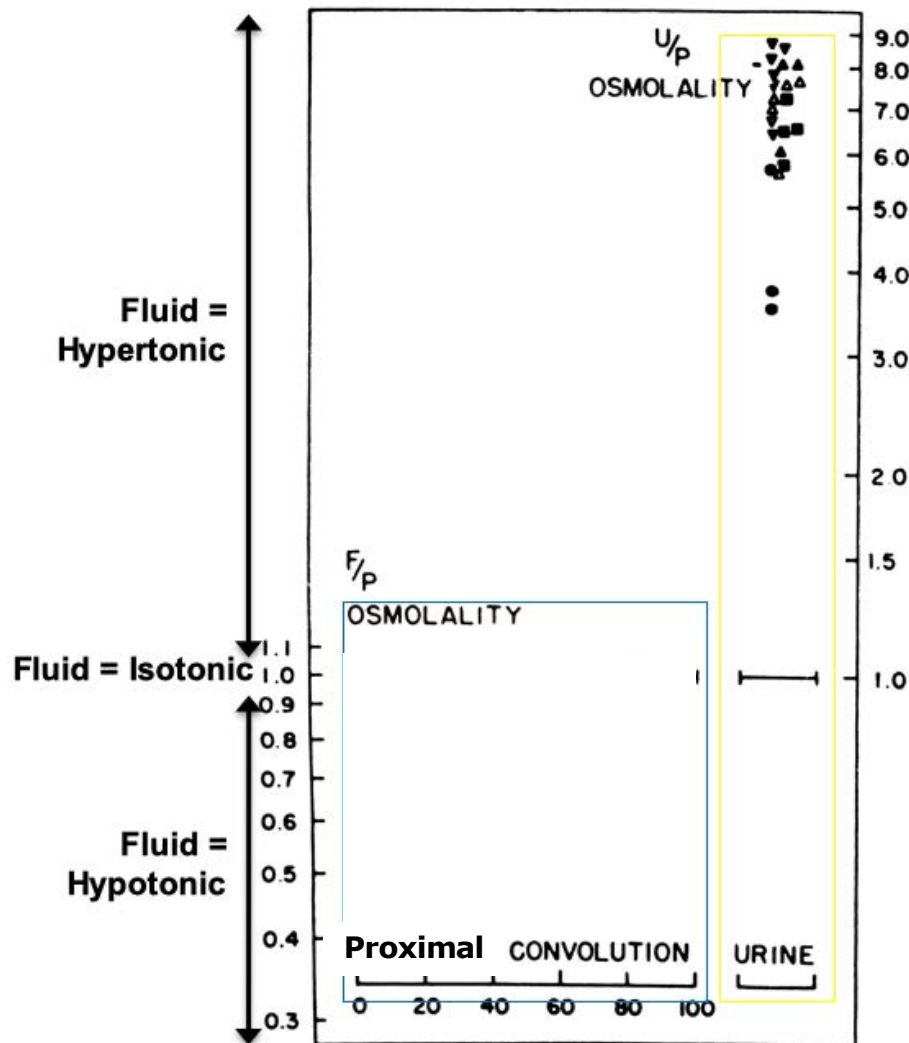


A Brief Preview of This Week

- Today
 - Review experimental results on urine concentration
 - Examine urinary system patient case studies
- Lab This Week
 - Urinary physiology activities
- Online Lecture Work This Week
 - A final look at kidney physiology!

The race to understand the kidneys

Is urine concentrated in the proximal tubule?

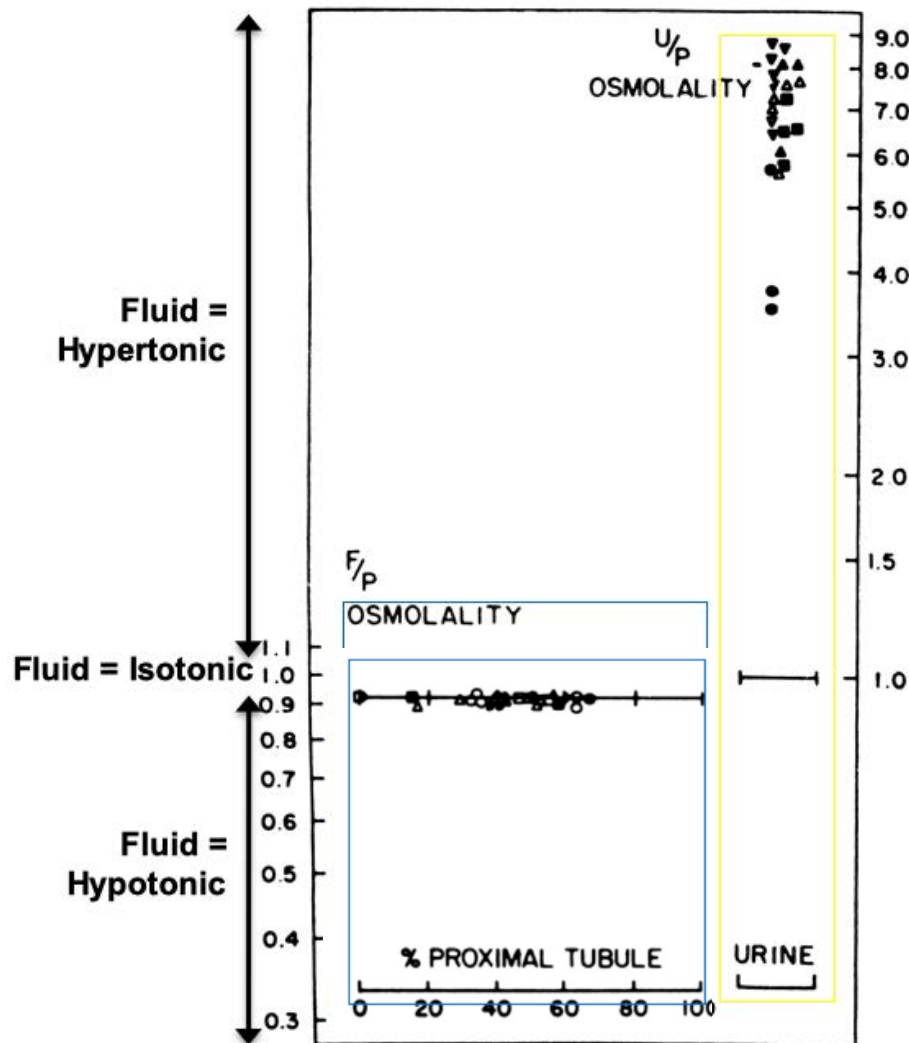


Make a rough graph like the one to the left on a piece of paper.

Place dots on the left half of the graph to indicate the proximal convolution data points you would expect to see if urine IS concentrated in the proximal tubule.

The race to understand the kidneys

Is urine concentrated in the proximal tubule?

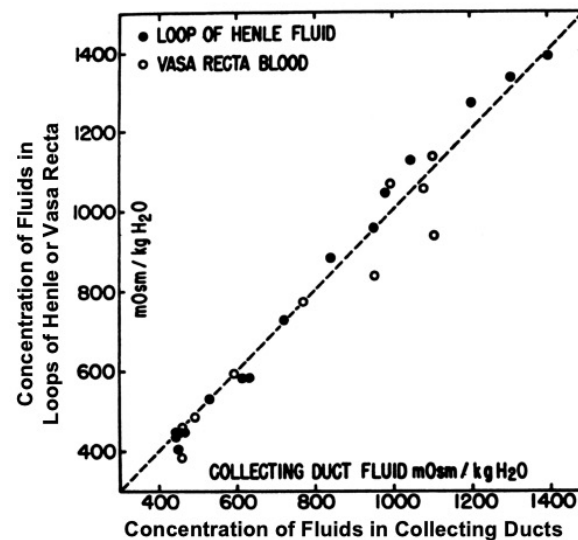
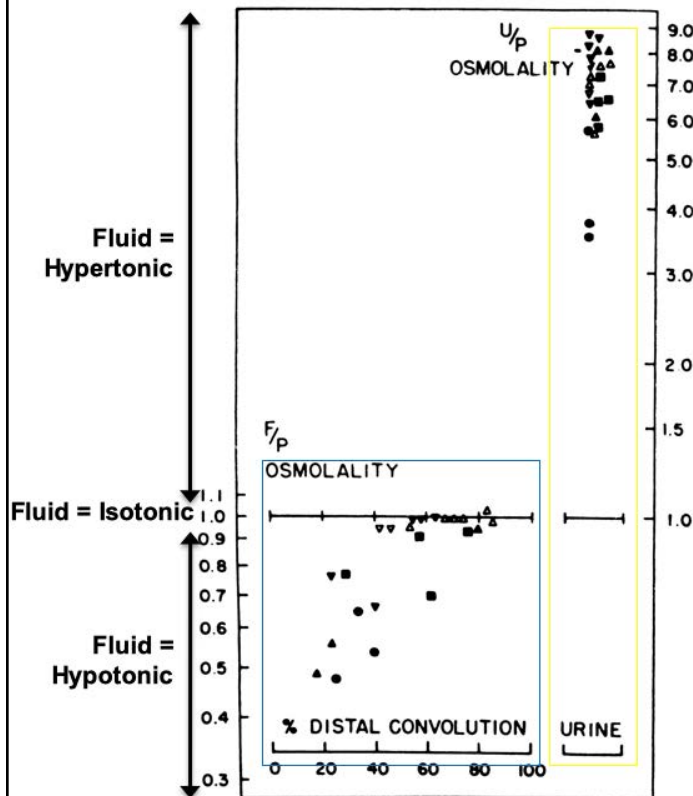


Make a rough graph like the one to the left on a piece of paper.

Place dots on the left half of the graph to indicate the proximal convolution data points you would expect to see if urine IS concentrated in the proximal tubule.

The race to understand the kidneys

Homer Smith hypothesized that the kidneys create concentrated (hypertonic) urine by *actively transporting water out of the distal tubule* and returning it to the blood. Margaret Mylle and her colleagues had an alternate hypothesis that *water returns to the blood via osmosis toward a salty area of the kidney*. Below are two key graphs showing data that evaluate these hypotheses.



Use evidence from these graphs to draw a conclusion about whose hypothesis is best supported. How do the graphs support your conclusions?



What will you be able to do today?

After class, you will be able to:

- Demonstrate an A&P mindset by adopting a cellular/molecular perspective on disease
- Explore important physiological activities in the nephron in the context of disease scenarios
- Identify and synthesize relevant physiology information from online sources



Case 1: Analysia

Analysia is a 52 year old woman coming into your clinic for a yearly physical. In the course of doing your checkup, you ask Analysia if anything has changed about her body or health since you last spoke.

Analysia remarks, “Mostly I’m doing pretty well! I’ve been trying to take care of myself by getting in some exercise and watching my diet, just like you advised me last time. I have to admit, though, that I find myself needing to use the restroom quite a lot these days. Maybe I’m just getting to be an old lady. I also have occasional, mild headaches, but again, I don’t think I can complain too much.”

**What do you think is important to note so far about
Analysia’s case?**

**What, if anything, do you already know that might be
relevant to this case?**



Case 1: Analysia

Analysia is a 52 year old woman coming into your clinic for a yearly physical. In the course of doing your checkup, you ask Analysia if anything has changed about her body or health since you last spoke.

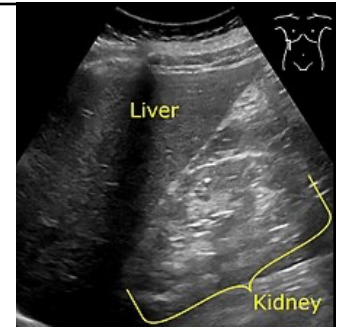
Analysia remarks, “Mostly I’m doing pretty well! I’ve been trying to take care of myself by getting in some exercise and watching my diet, just like you advised me last time. I have to admit, though, that I find myself needing to use the restroom quite a lot these days. Maybe I’m just getting to be an old lady. I also have occasional, mild headaches, but again, I don’t think I can complain too much.”

Step 1: Initial Information Gathering

Perform some research to gather information that might explain Analysia’s condition. Record any relevant findings.

- What are some different conditions that Analysia might be suffering from?
- What additional information might help lead you to a diagnosis?

Case 1: Analysisia



Analysisia remarks, “Mostly I’m doing pretty well! I’ve been trying to take care of myself by getting in some exercise and watching my diet, just like you advised me last time. I have to admit, though, that I find myself needing to use the restroom quite a lot these days. Maybe I’m just getting to be an old lady. I also have occasional, mild headaches, but again, I don’t think I can complain too much.”

You look at Analysisia’s record and recall that, 5 years ago, she was diagnosed with diabetes and has been in treatment for that condition. That leads you to perform some tests. You also notice that Analysisia’s legs appear puffy.

You receive the following results for Analysisia:

Test	Analysisia’s Results
Urinalysis	Albumin: 100 mg/24hr
Blood Creatinine	eGFR of 60 ml/min
Ultrasound	See picture
Blood Pressure	High

Step 2: Diagnosis

Perform additional research to interpret these results.

- What is being measured in the urinalysis and blood tests?
- What condition does Analysisia have?
- How do the urinalysis, blood, and ultrasound results relate to Analysisia’s condition?